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Real-Time Oriented Fabric Defect Detection via Position-Guided Fusion and Multiscale Rotation-Guided Strip Coupling

Huiyan Wang, *Member, IEEE*, Bingjie Chen, Chenhao Xie, and Wei Ye

***Abstract*—Real-time fabric inspection is essential for industrial manufacturing, where defects are often weak, slender, low-contrast, and arbitrarily oriented against repetitive textile backgrounds. Horizontal-box detectors introduce redundant background for tilted defects, while generic rotated detectors often lack textile-specific texture suppression and anisotropic modeling needed for accurate real-time inspection. To address these challenges, we formulate fabric inspection as a supervised rotated object detection task and propose a lightweight real-time rotated defect detection network (RRNet) tailored to industrial fabric scenes. RRNet introduces two task-specific modules: the position-guided fusion module, which performs fusion-stage positional-correspondence recalibration to preserve defect-consistent spatial evidence after cross-scale aggregation, and the multiscale rotation-guided strip coupling, which decomposes prediction-head modeling into multiscale anisotropic structure estimation and direction-conditioned modulation for structure–direction consistent prediction. In addition, we construct the industrial fabric defect (IFD) dataset with rotated annotations from real production lines to support realistic oriented defect detection. Experiments on IFD and the public ZJU-Leaper dataset under a unified oriented bounding box protocol demonstrate that RRNet achieves a favorable accuracy–efficiency tradeoff. Compared with the adopted deployment-oriented baseline, RRNet improves mAP@0.5 by 5.8% and 6.9% on IFD and ZJU-Leaper, respectively, while maintaining 140 frames per second (FPS) detector-side throughput.**

***Index Terms*—Fabric defect detection, position-guided fusion, real-time industrial inspection, rotated object detection, rotated strip convolution.**

I. Introduction

INDUSTRIAL fabric manufacturing requires reliable surface inspection, since even subtle defects can degrade product quality, disrupt downstream processing, and increase rework costs. Although imaging devices are widely used on production lines, many factories still rely on manual inspection, which is labor-intensive, subjective, and difficult to sustain at high line speeds. In practical line-scan systems, detectors must process many image patches per second and provide timely localization for online alarming and process feedback [1], [2], [3]. Therefore, fabric defect inspection requires detectors that are both accurate and deployment-friendly.

This task remains challenging in real factory environments. Fabric defects are often slender, low-contrast, and arbitrarily oriented, with large variations in scale, thickness, and aspect ratio. They also appear on repetitive textile backgrounds and are affected by folds, creases, illumination changes, and acquisition noise [1], [3], [4]. Consequently, weak defect cues can be overwhelmed by background texture, while isotropic feature modeling is insufficient for elongated and skewed defects. A practical method must therefore suppress texture interference, model anisotropic geometry, and satisfy real-time deployment constraints. For industrial fabric inspection, the desired output is not only defect presence but also fast instance-level localization with interpretable geometry. Horizontal-box detection is efficient, but axis-aligned boxes poorly fit elongated and tilted defects and introduce redundant background, as shown in Fig. 1(c). Anomaly detection reduces annotation cost, but it does not naturally provide category-aware and orientation-aware defect instances. Segmentation methods can recover fine contours, but their pixel-level outputs are less aligned with real-time instance-level localization. Therefore, we formulate fabric inspection as a supervised rotated object detection problem, where each defect is represented by an oriented bounding box (OBB) $(x, y, w, h,$ and $\theta)$. This formulation balances localization accuracy, geometric fidelity, and deployment efficiency.

Although rotated object detection has advanced significantly in domains, such as remote sensing and scene text detection [5], [6], [7], [8], [9], fabric inspection poses distinct challenges due to repetitive textile textures, slender low-contrast defects, and strict deployment requirements. However, most existing rotated detectors are designed for general oriented objects, and their generic fusion and prediction designs are less suited to suppressing texture interference and modeling weak anisotropic defect geometry in real-time fabric inspection. These limitations motivate a task-specific rotated detector for industrial fabric inspection.

To address these challenges, we propose a real-time rotated defect detection network tailored to industrial fabric defects

Received 28 May 2026; revised 7 July 2026; accepted 22 July 2026. This work was supported by the Key Research and Development Program of Zhejiang Province under Grant 2024C03172. Paper no. TII-26-4977. *(Corresponding author: Huiyan Wang.)*

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Digital Object Identifier 10.1109/TII.2026.3717200

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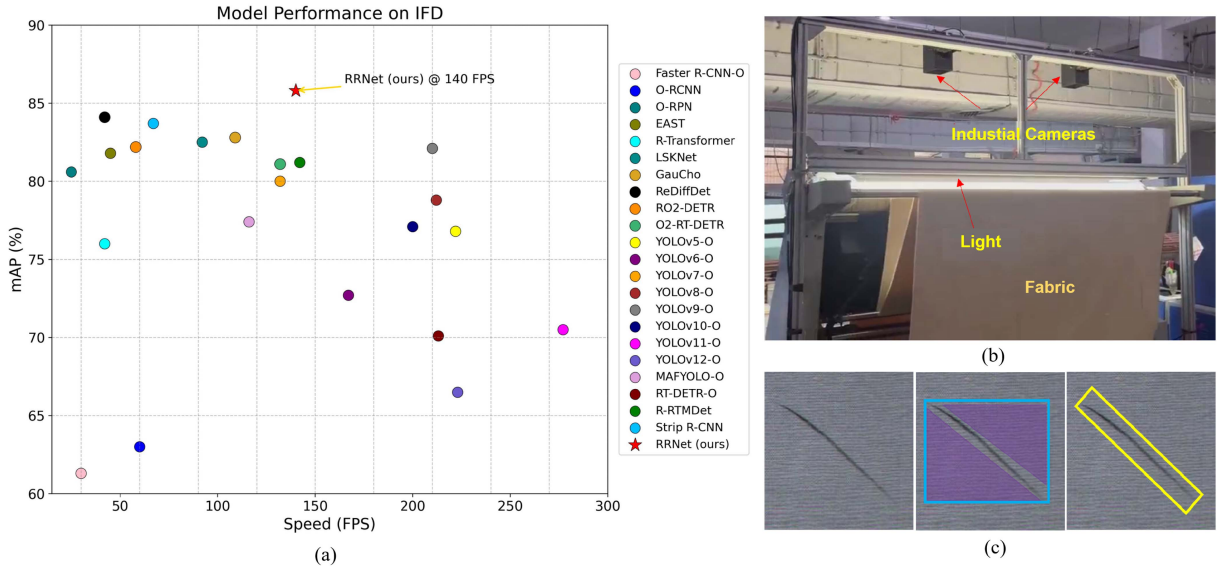


Fig. 1. Accuracy–throughput comparison, online inspection scenario, and annotation comparison for IFD detection. (a) Accuracy–throughput comparison on the IFD dataset, where RRNet achieves the best accuracy–efficiency tradeoff among the compared methods. (b) Line-scan inspection system for online fabric defect detection. (c) Comparison between horizontal and rotated bounding-box annotations. Horizontal boxes contain substantial redundant background for elongated and tilted defects, whereas rotated boxes provide tighter geometric localization.

(IFD). RRNet is built around two task-specific design principles. The first is fusion-stage positional correspondence preservation, implemented by the position-guided fusion module (PGFM). In textile scenes, weak defect evidence is particularly vulnerable when semantically strong but spatially coarse features are fused with higher resolution texture-rich features. PGFM is therefore introduced after cross-scale feature merging to recalibrate the fused representation itself. Different from generic coordinate-aware attention applied to a single feature stream, PGFM targets fusion-induced positional corruption: it preserves defect-consistent spatial evidence and suppresses texture-dominated responses before prediction, so that the subsequent detector operates on spatially more trustworthy features.

The second principle is structure–direction coupled head modeling, implemented by the multi-scale rotation-guided strip coupling (MSRSC). Even after feature fusion, the prediction head must handle large variations in defect scale, thickness, aspect ratio, and local orientation. MSRSC assigns complementary roles to two lightweight branches: the strip branch captures multiscale anisotropic structures, while the rotated branch provides feature-conditioned directional modulation. Their multiplicative coupling suppresses responses that are only structurally elongated or only directionally plausible, which is important for repetitive yarn-like backgrounds. Thus, RRNet emphasizes stage-specific structure–direction coupling rather than stacking high-capacity rotated operators.

To support realistic training and evaluation, we further construct the industrial fabric defect (IFD) dataset with rotated bounding-box annotations. Existing public datasets remain limited in industrial realism, oriented annotation, and defect diversity under complex textile textures. IFD contains 6332 images and 8242 defect instances from seven common defect categories collected in real production environments. It therefore provides a practical benchmark for category-aware rotated fabric defect detection. As illustrated in Fig. 1, RRNet achieves the highest detection accuracy and the most favorable accuracy–efficiency tradeoff on this benchmark.

The main contributions of this work are summarized as follows.

1) We formulate industrial fabric inspection as a supervised rotated object detection problem and develop RRNet, a deployment-friendly framework for accurate localization of slender and arbitrarily oriented defects.
2) We propose PGFM, a fusion-stage position-guided recalibration module that preserves target-consistent spatial correspondence after cross-scale aggregation and suppresses texture-dominated interference in repetitive textile backgrounds. Unlike generic feature reweighting, PGFM focuses on correcting fusion-induced positional corruption.
3) We propose MSRSC, a lightweight prediction-head module that couples multiscale anisotropic structure modeling with feature-conditioned direction modulation through multiplicative gating, improving the detection of elongated and skewed defects.
4) We construct the IFD dataset, a rotated-box annotated benchmark collected from real production environments, and validate RRNet on both IFD and ZJU-Leaper 37 under a unified rotated-detection protocol.

II. Related Work

A. Vision-Based Surface Defect Detection

Vision-based surface defect inspection mainly includes supervised detection, anomaly detection, and segmentation-based

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methods. Supervised detectors learn category-aware localization from labeled data and have achieved strong performance using convolutional backbones, multiscale fusion, deformable convolution, and transformer-based modeling [2], [10], [11], [12]. However, most existing detectors rely on horizontal boxes or isotropic feature modeling, which is less suitable for slender, tilted, and low-contrast fabric defects embedded in repetitive textile textures.

Unsupervised anomaly detection methods [13], [14], [15] reduce annotation cost by modeling normal patterns, but fabric folds, illumination changes, and texture irregularities may cause false alarms, and these methods do not naturally provide category- and orientation-aware instance localization. Segmentation methods can recover fine contours, but their pixel-level outputs are less aligned with high-throughput inspection, where fast instance-level localization and oriented geometric measurement are preferred. Therefore, this work focuses on supervised rotated detection for real-time fabric inspection, aiming to provide category-aware, orientation-aware, and deployment-friendly defect localization.

B. Rotated Object Detection

Rotated object detection has been widely studied in scene text detection, remote sensing, and ship detection, where objects are often dense, small, and arbitrarily oriented. In this work, we compare two groups of methods: native rotated detectors and general detector families evaluated under a unified OBB protocol.

Native rotated detectors, including rotation region proposal network (RRPN) [16], region of interest (RoI) Transformer [17], Oriented region-based convolutional neural network (R-CNN) [6], large selective kernel network (LSKNet) [7], GauCho [18], ReDiffDet [8], RO2-DETR [9], O2-RT-DETR [19], and Strip R-CNN [20], improve oriented localization through rotated proposals, feature alignment, large-kernel modeling, diffusion-based refinement, transformer-based detection, or strip-like structure modeling. General detector families under the unified OBB protocol, including models in the you only look once (YOLO) family [21], [22], [23], [24], [25], [26], [27], [28], MAF-YOLO [29], RT-DETR [30], and RTMDet [31], are included as deployment-oriented baselines.

However, neither group is fully tailored to industrial fabric inspection. Native rotated detectors mainly target general oriented objects and pay limited attention to repetitive textile textures, weak defect boundaries, and extreme aspect ratios. Unified-OBB general detectors are efficient, but their generic fusion and prediction heads lack explicit texture suppression and direction-sensitive modeling for slender fabric defects. In contrast, RRNet addresses these fabric-specific challenges through cross-scale positional preservation and structure–direction coupled prediction, achieving accurate rotated localization with a favorable accuracy–efficiency tradeoff.

III. Proposed Method

Fig. 2 illustrates the overall architecture of RRNet. RRNet is built on an efficient one-stage rotated detector based on efficient layer aggregation network (ELAN) architecture, which offers a favorable accuracy–efficiency tradeoff, stable multiscale feature aggregation, and a direct oriented detection pipeline for online localization [23]. Rather than redesigning the entire detector, RRNet addresses two fabric-specific bottlenecks: weak defect cues can be corrupted by repetitive texture responses during cross-scale fusion, and conventional isotropic prediction heads are poorly matched to slender and arbitrarily oriented defects.

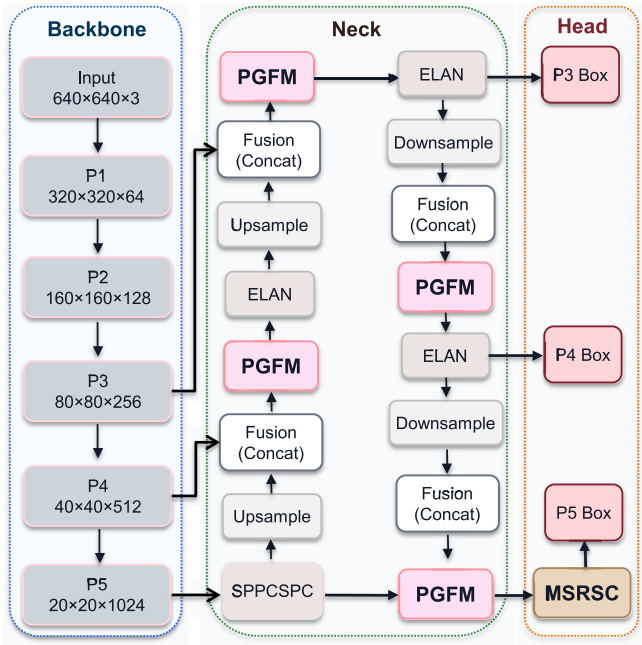


Fig. 2. Overall architecture of RRNet. PGFM preserves defect-consistent spatial evidence through postfusion recalibration at selected neck nodes, while MSRSC strengthens the P5 branch with structure–direction coupling for rotated defect prediction.

We adopt the anchor-based YOLOv7-O as the baseline, since its anchor priors better fit the extreme aspect ratios of slender oriented defects and its ELAN design provides clean module-insertion points.

Accordingly, RRNet enhances the neck and prediction head in a stage-specific manner. In the neck, PGFM is inserted at selected fusion nodes $\mathcal{S}_f$ to preserve defect-consistent spatial evidence after cross-scale aggregation. In the head, MSRSC replaces the original block on selected prediction levels $\mathcal{S}_h$ to strengthen anisotropic and orientation-aware prediction. In the default setting, PGFM is applied only at selected fusion locations and $\mathcal{S}_h = \{P5\}$, confining the extra computation to the most influential stages. Thus, PGFM improves the reliability of fused spatial evidence, while MSRSC converts that evidence into scale- and direction-consistent prediction responses. The complete inference flow is summarized in Table I. *Task formulation:* RRNet is formulated as a single-image supervised instance-level rotated object detector. Given an input fabric image I, the detector predicts a set of category-aware rotated instances

$$\hat{\mathcal{Y}} = f_\Theta(I) = \left\{(\hat{b}_i, \hat{c}_i, \hat{s}_i)\right\}_{i=1}^{N} \tag{1}$$

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TABLE I
INFERENCE FLOW OF RRNET

Input: Fabric image I
Output: Rotated defect instances $\hat{\mathcal{Y}}$

1: Input preprocessing
Normalize and resize I to obtain $\tilde{I}$.
2: Backbone feature extraction
Feed $\tilde{I}$ into the backbone to extract hierarchical features.
3: Neck fusion with PGFM
Perform multi-scale neck aggregation. For each selected fusion node $n \in \mathcal{S}_f$, take the post-fusion feature z_n and apply PGFM:
decompose z_n into directional branches according to Eq. (3);
generate axis-aware positional gates according to Eqs. (4)–(5);
perform residual directional calibration according to Eq. (6).
The selectively recalibrated neck feature $\tilde{z}_n$ is obtained by Eq. (7).
4: Detection pyramid construction
Generate multi-scale detection features $\mathcal{P} = \{P3, P4, P5\}$ from the PGFM-enhanced neck.
5: Prediction-head enhancement with MSRSC
For each selected prediction level $l \in \mathcal{S}_h$, apply MSRSC to the detection feature F_l:
compute anisotropic structural evidence $X_s^{(l)}$ according to Eqs. (8)–(9);
compute direction-conditioned modulation $X_r^{(l)}$ according to Eq. (10);
obtain the structure–direction coupled response $X_m^{(l)}$ according to Eq. (11).
The selectively enhanced prediction feature $\tilde{P}_l$ is obtained by Eq. (12). In the default setting, $\mathcal{S}_h = \{P5\}$, yielding prediction features $\{P3, P4, \tilde{P}_5\}$.
6: Rotated prediction and post-processing
Predict class scores and rotated boxes from the multi-scale features, decode (x, y, w, h, θ), and apply rotated Non-Maximum Suppression (NMS).
7: Return
Output the final rotated defect instances $\hat{\mathcal{Y}}$.

where Θ denotes the learnable parameters of RRNet, $\hat{b}_i = (\hat{x}_i, \hat{y}_i, \hat{w}_i, \hat{h}_i, \hat{\theta}_i)$ is the predicted rotated box, $\hat{c}_i$ is the predicted defect category, and $\hat{s}_i$ is the associated confidence score.

Training is supervised by rotated bounding-box annotations on the IFD dataset. Let the training set be $\mathcal{D} = \{(I_j, \mathcal{Y}_j)\}_{j=1}^{M}$, where $\mathcal{Y}_j$ denotes the rotated ground-truth instances of image I_j. Unless otherwise specified, RRNet follows the same sample assignment, prediction decoding, and rotated nonmaximum suppression protocol as the baseline detector, so that the gains can be attributed to feature propagation and head modeling rather than modified supervision or postprocessing. The overall optimization objective is written as

$$\mathcal{L} = \mathcal{L}_{\text{det}}^{\text{cls}} + \lambda_r \mathcal{L}_{\text{det}}^{\text{rbox}} \tag{2}$$

where $\mathcal{L}_{\text{det}}^{\text{cls}}$ denotes the baseline classification-related detection loss, $\mathcal{L}_{\text{det}}^{\text{rbox}}$ denotes the rotated localization loss defined on (x, y, w, h, θ), and λ_r is the balancing coefficient.

A. Position-Guided Fusion Module

PGFM addresses fusion-induced positional corruption during cross-scale aggregation. In fabric inspection, weak defect cues may be disturbed when spatially coarse semantic features are fused with high-resolution texture-rich features, producing responses that are semantically stronger but spatially less reliable. Therefore, PGFM is designed as a fusion-stage positional-correspondence recalibration operator rather than a generic attention block. Its key distinction lies in the postfusion placement: PGFM recalibrates the fused representation z_n, where positional corruption may already exist, instead of reweighting an isolated feature stream. This design preserves defect-consistent spatial evidence for subsequent rotated localization.

Let $\mathcal{S}_f$ denote the set of selected fusion nodes in the neck where PGFM is inserted. For a fusion node $n \in \mathcal{S}_f$, let $z_n \in \mathbb{R}^{H_n \times W_n \times C_n}$ denote the fused feature before recalibration. PGFM first decomposes z_n into two complementary directional branches

$$[f_n^x, f_n^y] = \text{Split}(z_n) \tag{3}$$

where the split is performed along the channel dimension. This keeps the original spatial lattice unchanged while separately modeling horizontal and vertical positional dependencies, which is suitable for elongated defect evidence. This decomposition prepares axis-aware positional gate generation rather than producing an unordered channel-attention vector. To encode long-range directional context without collapsing the full spatial coordinate system, 1-D pooling is applied to each branch to obtain axis-aware positional descriptors

$$u_n^x = \text{Pool}_H(f_n^x), \qquad u_n^y = \text{Pool}_V(f_n^y) \tag{4}$$

where $\text{Pool}_H(\cdot)$ and $\text{Pool}_V(\cdot)$ denote horizontal and vertical pooling, respectively. Unlike conventional 2-D global pooling, each operation aggregates context along only one axis while preserving coordinate information along the orthogonal axis. As a result, the pooled descriptors retain sensitivity to the subtle spatial irregularities that are critical for weak defect localization.

The pooled descriptors are then transformed into axis-aware directional positional gates

$$a_n^x = \sigma(\mathcal{T}_x(u_n^x)) \qquad a_n^y = \sigma(\mathcal{T}_y(u_n^y)) \tag{5}$$

where $\mathcal{T}_x(\cdot)$ and $\mathcal{T}_y(\cdot)$ denote lightweight transformations used to generate branch-specific position weights, and $\sigma(\cdot)$ is the sigmoid function. These gates are broadcast back to the spatial resolution of the corresponding branches and used for residual directional calibration

$$\hat{z}_n = \text{Concat}(f_n^x \odot (1 + a_n^x),\ f_n^y \odot (1 + a_n^y)) \tag{6}$$

where $\odot$ denotes elementwise multiplication and $\text{Concat}(\cdot)$ denotes channelwise concatenation. The residual form $(1 + a_n^{\cdot})$ is deliberate: the original local defect evidence is preserved explicitly, while the directional gate injects long-range context to correct fusion-induced positional corruption. Consequently, PGFM does not merely reweight saliency; it preserves defect-consistent spatial evidence during cross-scale aggregation so that subsequent stages operate on spatially more reliable responses.

For completeness, the neck output after selective insertion can be written as

$$\tilde{z}_n = \begin{cases} \hat{z}_n, & n \in \mathcal{S}_f \\ z_n, & \text{otherwise.} \end{cases} \tag{7}$$

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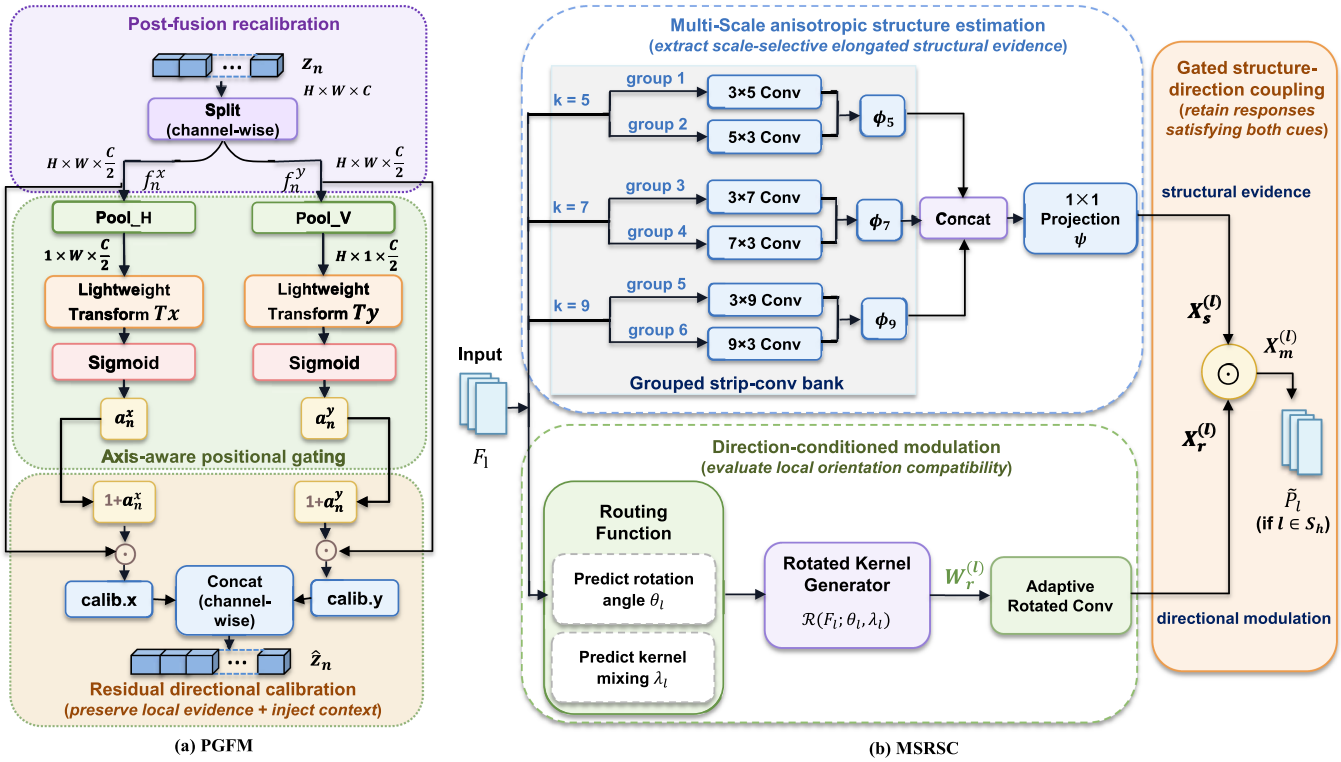


Fig. 3. Architectures of PGFM and MSRSC. (a) PGFM recalibrates postfusion features to preserve defect-consistent spatial evidence after cross-scale aggregation. (b) MSRSC separates anisotropic structure estimation from direction-conditioned modulation and couples them through multiplicative gating for structure–direction consistent rotated prediction.

This formulation highlights that PGFM is a *selective* fusion-stage enhancement rather than a uniform capacity increase over the whole pyramid.

Although PGFM, coordinate attention [32], and strip pooling [33] all use axiswise context, PGFM serves a different purpose: it recalibrates the postfusion feature z_n to mitigate fusion-induced positional corruption, applies axis-decoupled transformations to horizontal and vertical substreams, and uses residual gating to preserve local defect evidence during calibration.

B. Multiscale Rotation-Guided Strip Coupling

Although PGFM improves the reliability of postfusion spatial evidence, the prediction head still requires a geometric inductive bias matched to fabric defects. Standard square convolutions aggregate context isotropically, which may mix slender defect responses with nearby repetitive textile patterns and weaken rotated localization. MSRSC is therefore introduced to address the prediction-stage mismatch between isotropic head modeling and anisotropic, scale-varying defect geometry under a limited computation budget, rather than as a generic head enhancement block.

As shown in Fig. 3, MSRSC decomposes high-level head modeling into two role-separated branches. The multiscale strip branch estimates anisotropic structural evidence at different elongated scales, while the adaptive rotated-convolution branch provides direction-conditioned modulation by evaluating local orientation compatibility. Their multiplicative coupling retains responses only when elongated structure and directional consistency are jointly supported, suppressing yarn-like background patterns that activate only one cue. Thus, MSRSC differs from simply stacking strip convolution or adaptive rotated convolution; it performs prediction-stage structure–direction coupling for slender rotated fabric defects.

Let $\mathcal{S}_h$ denote the set of prediction levels where MSRSC is applied. For an input detection feature $F_l \in \mathbb{R}^{H_l \times W_l \times C_l}$ at level $l \in \mathcal{S}_h$, the strip branch first performs grouped multiscale anisotropic filtering. Let $\mathcal{K} = \{5, 7, 9\}$ denote the set of long-side strip kernel sizes, let $s = 3$ denote the fixed short-side size, and let g be the group number. The k-scale strip response is defined as

$$\phi_k(F_l) = \mathrm{Concat}\left(\mathrm{Conv}^{(g)}_{s\times k}(F_l),\ \mathrm{Conv}^{(g)}_{k\times s}(F_l)\right), k \in \mathcal{K} \quad (8)$$

where $\mathrm{Conv}^{(g)}_{s\times k}(\cdot)$ and $\mathrm{Conv}^{(g)}_{k\times s}(\cdot)$ denote grouped anisotropic strip-like convolutions along two orthogonal directions. Compared with isotropic large-kernel aggregation, this branch enlarges the receptive field mainly along dominant elongated axes and is therefore better aligned with the morphology of slender fabric defects. The fixed short side s further preserves a small local 2-D neighborhood, which helps retain finite-width defect boundaries and local shape continuity while avoiding excessive integration of surrounding repetitive textures.

We set the strip short side to $s = 3$, which better preserves defect-boundary continuity and suppresses single-line texture

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responses than $s = 1$ under periodic yarn textures, while avoiding the excessive background context introduced by $s = 5$.

The strip responses at different receptive scales are then aggregated as

$$X_s^{(l)} = \psi(\text{Concat}(\{\phi_k(F_l)\}_{k\in\mathcal{K}})) \tag{9}$$

where $\psi(\cdot)$ denotes a lightweight 1×1 projection used to fuse multiscale and multidirectional strip responses. In this way, $X_s^{(l)}$ represents anisotropic structural evidence that is selective to both elongated scale and dominant direction, while maintaining a limited computation budget through grouped filtering.

In parallel, the adaptive rotated-convolution branch generates an orientation-aware response from the same input. Following the routing strategy in [34], a routing function predicts the rotation parameter and kernel mixing coefficient, which are used to generate a feature-adaptive rotated kernel

$$W_r^{(l)} = \mathcal{R}(F_l; \theta_l, \lambda_l), \qquad X_r^{(l)} = \text{Conv2}d(F_l, W_r^{(l)}). \tag{10}$$

Unlike using adaptive rotated convolution as a standalone substitute for a standard kernel, RRNet uses this branch as a direction-conditioned modulation stream. Therefore, the rotated branch does not simply rotate filters; it evaluates whether the incoming high-level response is locally compatible with a plausible defect orientation.

The outputs of the two branches are fused by element- multiplication

$$X_m^{(l)} = X_s^{(l)} \odot X_r^{(l)} \tag{11}$$

where $\odot$ denotes elementwise multiplication. This fusion performs gated structure–direction coupling rather than heuristic branch accumulation. The strip response $X_s^{(l)}$ captures elongated structural evidence at multiple receptive scales, while the rotated response $X_r^{(l)}$ serves as an orientation-conditioned gate. Their elementwise product retains strong responses only when the elongated structure is also directionally consistent. This is important in textile images, where repetitive yarn-like textures may trigger either structural or directional responses alone; additive or concatenation-based fusion would preserve such partial evidence, whereas multiplication suppresses it. Therefore, MSRSC differs from simply appending strip convolution or adaptive rotated convolution as standalone head enhancers.

The selectively enhanced prediction feature can therefore be written as

$$\tilde{P}_l = \begin{cases} X_m^{(l)}, & l \in \mathcal{S}_h \\ F_l, & \text{otherwise.} \end{cases} \tag{12}$$

In the default RRNet, $\mathcal{S}_h = \{P5\}$, so the additional computation is confined to the high-level prediction stage, where semantic abstraction is strongest and orientation-aware anisotropic modeling yields the highest return under limited computational budget. This selective definition also makes clear that MSRSC replaces the original high-level head block on the chosen level(s) rather than being appended after it, which is consistent with the deployment-aware design objective of RRNet.

Compared with strip-only directional modules, adaptive rotated convolution alone, or generic large-kernel head blocks, MSRSC is not a sequential stack of existing operators. Instead, it decomposes high-level head modeling into two nonredundant roles: multiscale anisotropic structure estimation and direction-conditioned modulation. Their multiplication in (11), rather than simple accumulation, retains strong responses only when elongated structure and local direction are simultaneously consistent. This is particularly useful in textile images, where repetitive background textures may mimic elongated defect structures but fail to satisfy direction consistency. Therefore, MSRSC complements PGFM rather than overlapping with it: PGFM preserves defect-consistent spatial evidence during fusion, and MSRSC converts that evidence into scale- and direction-consistent high-level responses for final rotated detection.

MSRSC adopts multiplicative fusion as a joint-consistency gate. Unlike additive fusion, which may retain single-cue yarn textures or edge patterns, multiplicative fusion produces strong responses mainly when structural and directional cues agree, thereby suppressing background responses while preserving defect regions supported by both cues.

IV. Evaluation Dataset

A. IFD Dataset

Existing public datasets for fabric defect inspection are limited in industrial realism, oriented annotation, and coverage of elongated defects under complex textile textures [37]. To address this gap, we build the IFD dataset using the acquisition system shown in Fig. 1(b), which is deployed on a real production line.

In practice, RRNet is deployed on the production-line industrial workstation equipped with an RTX 3090 GPU. The online inspection system uses two 8 K line-scan cameras to cover a 1.6 mwide production line, with each camera observing a 0.8 m swath. The spatial resolution is 0.0977 mm/pixel, and the acquired streams are cropped into 640 × 640 patches. At a fabric speed of 60 m/min, full-width inspection produces about 416 patches/s, corresponding to about 208 patches/s per camera, i.e., about 4.81 ms per patch. RRNet achieves 140 frames per second (FPS) before deployment optimization, corresponding to about 7.14 ms per patch, and about 339 patches/s after the measured 2.42× TensorRT FP16 acceleration, corresponding to about 2.95 ms per patch. With camerawise parallel inference, the per-camera throughput requirement remains below the optimized detector throughput even with a 20% system-overhead margin. Therefore, RRNet is feasible for real-time on-site deployment at 60 m/min.

All images are annotated by experienced inspectors using rotated bounding boxes. The final dataset contains 6332 images and 8242 defect instances from seven common defect categories: hole (HO), hooked yarn (HY), broken ends (BE), broken picks (BP), linear oil (LO), scratch mark (SM), and dirty (DI).

IFD covers multiple cloth types with substantial variations in color, weave pattern, and texture. The seven categories exhibit distinct geometric and appearance properties. HY, BE, BP, and LO are typically elongated defects. BE is mostly near-vertical with limited orientation variation, whereas BP is generally near-horizontal but shows greater orientation variation than BE and is more prone to confusion with yarn-like background

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textures. HY and LO appear with more diverse orientations. HO is nearly isotropic and lacks stable orientation cues. DI and SM are strongly affected by texture confusion and scale variation. Therefore, IFD reflects the main challenges of IFD detection, including orientation variation, elongated structures, low contrast, repetitive texture interference, and large intraclass geometric variation.

In our deployment setting, the detector is trained to handle all cloth types on the same production line under fixed acquisition conditions and to generalize to unseen defect instances from the same production distribution. We therefore do not use a cloth-type held-out split; instead, test images are collected from different time periods of continuous production. To avoid information leakage while preserving the production distribution, we split the data at the raw-image level before patch generation and augmentation. Considering class and cloth-type imbalance, we adopt an 8:2 distribution-preserving split. Data augmentation is applied only to the training set.

B. ZJU-Leaper Dataset

To complement IFD with a public benchmark, we also evaluate on ZJU-Leaper [37]. It is a widely used fabric defect dataset containing 27 650 defective images and 71 127 defect-free images from 19 fabric pattern types, all at a resolution of 512×512. The original annotations provide horizontal boxes and pixel-level masks, and all defects are treated as a single class.

Since rotated boxes are not provided, we convert each defect mask into a minimum-area rotated rectangle and manually inspect degenerate cases. All methods are then trained and evaluated under the same one-class rotated-detection protocol to ensure a fair comparison with the rotated setting of this work.

IFD and ZJU-Leaper are evaluated independently under their respective acquisition conditions using a unified OBB protocol. Cross-dataset zero-shot transfer is not used because it would mainly test domain generalization across different imaging setups rather than the deployment setting considered in this work.

V. Experiment

A. Implementation Details

RRNet is implemented in PyTorch and trained on Ubuntu 22.04 with Python 3.9.15, PyTorch 1.10.0, and CUDA 11.3. All experiments are conducted on a workstation with an Intel Core i7-11700 CPU (2.50 GHz), 32 GB RAM, and an NVIDIA RTX 3090 GPU. Unless otherwise stated, all average precision (AP) metrics are reported at an IoU threshold of 0.5. On IFD, we report per-class AP@0.5 and overall mAP@0.5; on ZJU-Leaper, where all defects are treated as a single class, we report single-class mAP@0.5. Detector-side throughput is reported in FPS and measured on a single RTX 3090 after warm-up. Recall is additionally reported in the ablation study for a more complete evaluation of detection performance.

RRNet is trained for 300 epochs with Adam using an initial learning rate of 1×10^{-3}, $\beta_1 = 0.9$, weight decay of 5×10^{-4}, batch size of eight, and input size of 640×640. A three-epoch warm-up is used with warm-up momentum set to 0.8. For fair comparison, all methods use the same train/validation split, input resolution, data augmentation, rotated-box decoding, and rotated NMS unless otherwise stated. RRNet and the YOLOv7-based baseline are initialized with official YOLOv7 pretrained weights [23], while other methods follow their official pretrained initialization settings. For YOLO-family baselines, the medium-sized variant is used when available; otherwise, the closest parameter-budget model is selected and reported. Across all experiments, the best-performing results are highlighted in bold font, and the second-best results are formatted in italics.

B. Comparison With SOTA Models

Table II compares RRNet with native rotated detectors and general detector families under a unified OBB protocol. For compactness, this broad comparison reports category AP@0.5, overall mAP@0.5, mAP@0.5:0.95, model complexity, and detector-side FPS. RRNet achieves the best mAP@0.5 and mAP@0.5:0.95 on both IFD and ZJU-Leaper while maintaining 140 FPS, showing the most favorable accuracy–efficiency trade-off among the compared methods, as also illustrated in Fig. 1(a).

Compared with native rotated detectors, RRNet achieves higher accuracy with much better efficiency. ReDiffDet obtains the second-best mAP@0.5 on both datasets, but runs at only 42 FPS. In contrast, RRNet improves mAP@0.5 from 84.1% to 85.8% on IFD and from 62.7% to 66.3% on ZJU-Leaper, while being more than three times faster. Under the stricter mAP@0.5:0.95 metric, RRNet also outperforms ReDiffDet on both IFD and ZJU-Leaper by 2.6% and 2.8%, respectively, demonstrating that its advantage also holds under high-precision localization evaluation. Compared with O2-RTDETR, the fastest native rotated detector in Table II, RRNet improves mAP@0.5 by 4.7% on IFD and 7.3% on ZJU-Leaper with comparable throughput.

Among general detector families, RRNet also provides the strongest accuracy under practical speed. YOLOv11-O achieves the highest FPS, but its mAP@0.5 is much lower than RRNet on both datasets. More importantly, compared with the YOLOv7-O baseline, RRNet improves mAP@0.5 by 5.8% on IFD and 6.9% on ZJU-Leaper, and improves mAP@0.5:0.95 by 3.3% on IFD and 5.1% on ZJU-Leaper, while increasing FPS from 132 to 140. This indicates that the proposed modules substantially improve detection accuracy under both loose- and high-precision localization without sacrificing deployment efficiency.

The categorywise results further demonstrate RRNet's advantage on fabric-specific defects. On IFD, RRNet achieves the best AP in five of seven categories and improves six categories over YOLOv7-O, with the largest gain on BP (+18.6%). This agrees with the module design: PGFM strengthens fused spatial evidence under repetitive textures, while MSRSC enhances structure–direction consistent prediction for elongated defects. Although R-RTMDet obtains the best BE AP and YOLOv7-O is slightly better on HO, both have lower overall mAP@0.5 than RRNet. The HO result is consistent with its nearly isotropic shape and weak orientation cues. Overall, RRNet improves the dominant challenging categories while maintaining the best dataset-level performance.

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TABLE II
COMPARISON WITH NATIVE ROTATED DETECTORS AND UNIFIED-OBB DETECTOR-FAMILY BASELINES ON IFD AND ZJU-LEAPER

Model	Params (M)	FLOPs (G)	IFD										ZJU-Leaper	
			Category AP@0.5(%)							mAP@0.5 (%)	mAP@0.5: 0.95(%)	FPS	mAP@0.5 (%)	mAP@0.5: 0.95(%)
			HY	DI	BE	BP	LO	SM	HO					
Native rotated object detectors														
F-RCNN-O [35]	42.3	185.0	66.7	48.5	61.8	51.8	64.0	72.8	63.5	61.3	38.5	30	42.3	22.6
Oriented RCNN [6]	44.8	198.0	75.8	51.2	62.1	50.4	65.9	77.1	58.7	63.0	41.2	60	41.5	22.1
RRPN [16]	46.5	210.0	85.4	77.0	76.9	60.7	81.5	88.1	94.8	80.6	61.8	25	42.0	22.4
EAST [36]	13.2	45.0	85.0	79.6	76.9	65.0	80.5	90.3	95.4	81.8	62.4	45	50.5	27.7
RoI TransFormer [17]	53.6	228.0	79.0	75.8	73.2	53.2	70.6	87.8	92.6	76.0	57.7	42	48.5	26.9
LSKNet [7]	31.4	102.0	86.0	78.4	79.2	66.5	82.5	89.9	94.9	82.5	63.4	92	57.8	32.8
GauCho [18]	33.7	109.0	86.4	78.5	80.1	66.8	82.7	89.3	96.1	82.8	63.8	109	60.4	34.8
ReDiffDet [8]	49.8	216.0	88.0	80.8	79.4	70.9	84.3	89.8	95.2	84.1	65.3	42	62.7	36.4
RO2-DETR [9]	46.9	182.0	84.9	76.8	81.2	67.9	81.5	88.7	94.6	82.2	63.6	58	59.8	34.3
O2-RT-DETR [19]	25.6	68.9	84.5	76.1	79.3	64.8	80.8	88.0	94.5	81.1	62.1	132	59.0	33.8
Strip R-CNN [20]	34.5	118.0	85.5	81.5	77.8	73.0	82.9	90.9	94.3	83.7	65.0	67	60.6	35.1
General detector families under a unified OBB protocol														
YOLOv5-O [21]	21.4	50.5	83.3	74.6	65.9	51.0	79.5	88.9	94.8	76.8	59.5	222	48.7	27.0
YOLOv6-O [22]	34.8	85.5	79.3	72.1	63.0	48.1	69.9	83.1	93.3	72.7	55.0	167	46.2	25.4
YOLOv8-O [24]	26.0	79.0	84.1	79.2	65.6	56.2	81.1	89.5	95.8	78.8	61.2	212	55.6	31.9
YOLOv9-O [25]	25.3	77.5	85.9	81.3	80.5	62.8	82.4	87.2	95.0	82.1	63.5	210	53.2	30.1
YOLOv10-O [26]	16.1	59.0	83.8	75.9	64.7	54.7	79.7	86.8	93.8	77.1	59.7	200	52.6	29.8
YOLOv11-O [27]	20.2	68.0	79.2	68.7	51.9	42.7	72.7	83.8	94.7	70.5	53.7	**277**	58.7	33.2
YOLOv12-O [28]	20.8	71.0	75.3	65.8	58.5	20.5	75.5	81.5	88.2	66.5	49.8	223	56.3	31.7
MAF-YOLO-O [29]	28.7	92.0	82.8	75.3	80.7	43.5	77.5	87.5	94.4	77.4	58.9	116	54.3	30.3
RT-DETR-O [30]	21.8	60.5	79.0	67.4	68.7	27.6	71.5	82.5	93.8	70.1	52.6	213	56.5	32.3
R-RTMDet [31]	27.8	81.0	80.2	74.0	**86.5**	71.3	79.3	87.0	89.8	81.2	62.8	142	57.8	32.7
YOLOv7-O (baseline) [23]	25.8	77.8	86.1	76.1	76.8	55.1	81.2	88.0	**96.3**	80.0	64.6	132	59.4	34.1
YOLOv7-O-Wide [23]	27.0	84.5	86.6	77.4	77.6	58.4	82.3	88.7	96.5	81.1	65.1	124	60.5	34.7
RRNet (Ours)	26.9	84.7	**89.2**(+3.1)	**83.4**(+7.3)	82.0(+5.2)	**73.7**(+18.6)	**86.0**(+4.8)	**91.2**(+3.2)	94.8(-1.5)	**85.8**(+5.8)	**67.9**(+3.3)	140(+8)	**66.3**(+6.9)	**39.2**(+5.1)

Parentheses indicate gains over the baseline.

TABLE III
ABLATION STUDY OF MSRSC AND PGFM ON IFD AND ZJU-LEAPER

MSRSC	PGFM	Params (M)	FLOPs (G)	IFD										ZJU-Leaper		
				Category AP@0.5(%)							Recall (%)	mAP@0.5 (%)	mAP@0.5: 0.95(%)	Recall (%)	mAP@0.5 (%)	mAP@0.5: 0.95(%)
				HY	DI	BE	BP	LO	SM	HO						
-	-	25.8	77.8	86.1	76.1	76.8	55.1	81.2	88.0	96.3	69.0	80.0	64.6	52.4	59.4	34.1
+	-	25.5	77.4	85.4	78.7	80.8	70.7	82.7	89.9	95.8	73.8	83.4	66.4	56.5	63.1	36.8
-	+	27.2	85.1	87.3	82.6	**84.2**	70.8	**86.0**	90.6	**96.8**	74.2	85.5	67.5	57.1	65.2	38.0
+	+	26.9	84.7	**89.2**	**83.4**	82.0	**73.7**	**86.0**	**91.2**	94.8	**75.3**(+6.3)	**85.8**(+5.8)	**67.9**(+3.3)	**58.4**(+6.0)	**66.3**(+6.9)	**39.2**(+5.1)

Parentheses indicate gains over the baseline.

On ZJU-Leaper, RRNet again achieves the best mAP@0.5 of 66.3% and the best mAP@0.5:0.95 of 39.2%, outperforming both native rotated detectors and unified-OBB detector-family baselines. Since ZJU-Leaper differs from IFD in data source, defect appearance, and class distribution, this result indicates that RRNet generalizes beyond the self-built dataset and remains effective for broader fabric defect inspection scenarios.

To decouple architectural gain from capacity, we compare RRNet with a width-scaled YOLOv7-O (YOLOv7-O-Wide) matched in Params/FLOPs (27.0 M / 84.5 G versus 26.9 M / 84.7 G). This baseline only marginally improves over YOLOv7-O (81.1%/65.1% versus 80.0%/64.6% on mAP@0.5/mAP@0.5:0.95) and remains clearly below RRNet (85.8%/67.9%). Other comparable- or larger capacity detectors in Table II also underperform RRNet, confirming that the gain is not due to capacity alone.

C. Ablation Study

We conduct ablation studies to examine three aspects: 1) the individual contributions of PGFM and MSRSC; 2) whether the two modules are complementary; and 3) whether their gains are beyond those of common lightweight alternatives. The main results on IFD and ZJU-Leaper are reported in Table III, and the detailed analyses are given in Tables IV–VIII.

Effectiveness of PGFM and MSRSC: Table III reports both detection performance and model complexity for each configuration. The full RRNet achieves the best overall performance on both datasets. Compared with the baseline, it improves Recall, mAP@0.5, and mAP@0.5:0.95 by 6.3%, 5.8%, and 3.3% on IFD, and by 6.0%, 6.9%, and 5.1% on ZJU-Leaper, respectively. These consistent gains demonstrate that improving both cross-scale fusion and prediction-head geometry is beneficial for rotated fabric defect detection. Meanwhile, the added complexity is limited. PGFM adds only 1.4 M Params and 7.3 G

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TABLE IV
COMPARISON OF PGFM WITH LIGHTWEIGHT ATTENTION ALTERNATIVES ON THE IFD DATASET

Method / Mechanism	Recall(%)	mAP@0.5(%)	FPS
Baseline	69.0	80.0	**132**
+SE [38] / channel reweighting	71.1	82.6	122
+CBAM-Spatial [39] / spatial reweighting	71.9	83.0	126
+CoordAttention [32] / coordinate-aware reweighting	72.6	83.8	124
+Strip Pooling [33] / strip context injection	72.2	83.4	121
+PGFM / post-fusion positional recalibration	**74.2**	**85.5**	130

The best-performing results are highlighted in bold, and the second-best results are underline.

TABLE V
ABLATION OF MSRSC ON THE IFD DATASET

Head design / Design factor	Recall(%)	mAP@0.5(%)	FPS
Baseline head block	69.0	80.0	132
Comparison with related head designs			
+LSK-like [7]	72.5	82.6	121
Internal analysis of MSRSC coupling			
+Strip-only [33]	71.4	81.2	164
+ARC-only [34]	71.9	81.8	154
+Strip+ARC (add)	72.9	82.7	166
+Strip+ARC (concat)	73.1	82.8	158
+Strip⊙ARC, constant gate	71.6	81.4	166
+Strip⊙ARC, shuffled gate	72.0	81.7	165
+MSRSC (Strip⊙ARC, spatially aligned)	**73.8**	**83.4**	**167**

The best-performing results are highlighted in bold, and the second-best results are underline.

TABLE VI
ABLATION ON MSRSC DESIGN CHOICES ON THE IFD DATASET

Fusion	Placement	Deploy	Recall(%)	mAP@0.5(%)
add	P5	replace	72.9	82.7
mul	**P5**	**replace**	**73.8**	**83.4**
mul	P5	append	73.5	83.0
mul	P3	replace	71.2	81.7
mul	P4	replace	72.5	82.6
mul	P4+P5	replace	73.7	83.2
mul	P3+P4+P5	replace	73.1	82.8

The best-performing results are highlighted in bold, and the second-best results are underline.

FLOPs, corresponding to 5.4% and 9.4% increases over the baseline, while MSRSC slightly reduces complexity (-0.3 M Params and -0.4 G FLOPs) by replacing the original head block. Overall, RRNet adds only 1.1 M Params and 6.9 G FLOPs while improving IFD mAP@0.5 by 5.8%.

PGFM provides broad fusion-level improvements. On IFD, PGFM alone raises Recall and mAP@0.5 by 5.2% and 5.5%, respectively, and improves all defect categories, especially DI, BE, BP, and LO. On ZJU-Leaper, it improves Recall and mAP@0.5 by 4.7% and 5.8%. These results indicate that PGFM effectively suppresses texture interference and preserves reliable spatial evidence during cross-scale fusion, benefiting a wide range of defect types.

MSRSC provides targeted prediction-level improvements. Used alone, it improves Recall and mAP@0.5 by 4.8% and 3.4% on IFD, and by 4.1% and 3.7% on ZJU-Leaper. Its strongest categorywise gain appears on BP, whose AP increases from 55.1% to 70.7%, followed by BE from 76.8% to 80.8%. This confirms that MSRSC is effective for elongated defects with informative direction cues by enforcing structure–direction consistency in the prediction head.

The complementarity of the two modules is most evident on BP. PGFM improves BP AP to 70.8% by providing cleaner fused evidence under repetitive textures, while MSRSC improves it to 70.7% by exploiting its largely horizontal elongated structure. Their combination further increases BP AP to 73.7%, showing that PGFM and MSRSC address different bottlenecks: fusion reliability and direction-consistent prediction.

The categorywise effects are not strictly additive for all classes, but they remain consistent with defect geometry. For BE, which is mostly near-vertical with limited orientation variation, PGFM already addresses the dominant bottleneck by improving fused spatial evidence, so additional direction gating provides limited new information. For HO, the nearly isotropic shape provides weak orientation cues, making structure–direction coupling less helpful. Overall, PGFM offers broad and stable fusion-level gains, while MSRSC provides targeted benefits for elongated and direction-informative defects; together, they deliver the strongest dataset-level performance.

Comparison with alternative module designs: Table IV compares PGFM with lightweight alternatives inserted at the same neck fusion stage. The squeeze-and-excitation (SE) module and the spatial branch of the convolutional block attention module (CBAM) perform generic channel or spatial reweighting, while CoordAttention and Strip Pooling introduce axiswise contextual modeling. PGFM achieves the best Recall and mAP@0.5 while maintaining 130 *FPS*, showing that postfusion positional recalibration is more effective than generic attention reweighting for preserving weak defect evidence. Table VII further ablates PGFM by reverting its postfusion placement, axis-decoupled transformation, and residual gating. Each variant reduces mAP@0.5 and mAP@0.5:0.95, confirming that these designs jointly contribute to the improvement.

Table V evaluates MSRSC through external head comparison and internal two-branch coupling analysis. Strip-only and only adaptive rotated convolution (denoted as ARC-only) improve the baseline but remain below MSRSC. MSRSC also outperforms the stronger LSK-like head with a 46 FPS speedup, achieving the best accuracy–efficiency tradeoff. For branch coupling, additive fusion (82.7% mAP@0.5) and concatenation fusion (82.8%), both underperform MSRSC (83.4%). Capacity-preserving variants that disrupt multiplicative gating, including constant gating (81.4%) and spatially shuffled rotated responses (81.7%), also drop to the single-branch level. These results show that the gain comes from spatially aligned, direction-conditioned multiplicative gating, rather than individual operators or added capacity. This supports separating anisotropic structure estimation from direction-conditioned modulation.

Table VI further analyzes the internal design of MSRSC. Multiplicative coupling outperforms additive fusion, suggesting that direction-aware responses are more effective when used to gate anisotropic structural evidence rather than simply accumulate with it. Replacing the selected head block performs better than appending an extra block, indicating that the gain is not mainly

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TABLE VII
ABLATION STUDY OF PGFM DESIGN CHOICES ON THE IFD DATASET

Variant	Recall(%)	mAP@0.5(%)	mAP@0.5:0.95(%)
Baseline	69.0	80.0	64.6
w/ pre-fusion placement	72.8	84.1	66.1
w/ shared axis-coupled transform	<u>73.1</u>	<u>84.5</u>	<u>66.5</u>
w/ pure multiplicative gate ($f \odot a$)	72.4	83.7	65.8
Full PGFM	**74.2**	**85.5**	**67.5**

The best-performing results are highlighted in bold, and the second-best results are underline.

TABLE VIII
EFFECT OF THE STRIP SHORT SIDE IN MSRSC ON IFD

Short side s	Recall(%)	mAP@0.5(%)	mAP@0.5:0.95(%)
$s = 1$	72.7	82.6	65.3
$s = 3$ **(ours)**	**73.8**	**83.4**	**66.4**
$s = 5$	73.1	82.7	65.8

caused by increased capacity. The P5-only setting also achieves the best result, showing that high-level features provide the most suitable semantic abstraction for structure–direction coupling under a limited computation budget.

Effect of the strip short side: As shown in Table VIII, performance peaks at $s = 3$ (83.4% mAP@0.5 and 66.4% mAP@0.5:0.95) and decreases with both smaller $s = 1$ and larger $s = 5$ short sides, suggesting that a moderate strip width best balances local detail and contextual coverage.

D. Qualitative Analysis

Fig. 4 shows representative detection examples on the IFD dataset. We compare RRNet with the baseline [23] and GauCho [18] in complex scenarios (Rows 1–3) and representative failure cases (Rows 4 and 5).

For complex scenarios involving multiple defects and cluttered backgrounds, RRNet detects more defect instances and provides tighter rotated localization, while the baseline tends to miss weak or slender defects. In addition, both the baseline and GauCho may confuse visually similar categories under repetitive textile backgrounds.

For the failure cases, Row 4 shows two defects embedded in heavy wrinkle texture. Both the baseline and GauCho miss the two ground-truth defects and produce only false detections. RRNet detects both defects with correct categories, but partially truncates the elongated defect, indicating remaining difficulty under strong background interference. Row 5 shows an extremely low-contrast defect on dark fabric. Although all three methods miss it, RRNet produces fewer false alarms than the baseline and GauCho. These cases show that heavily textured backgrounds and extremely low-contrast dark-surface defects remain challenging.

Fig. 5 presents Grad-CAM [40] visualizations of representative elongated defects. RRNet is compared with the YOLOv7-O baseline, YOLOv9-O, and GauCho. Compared with these methods, RRNet yields more continuous defect-aligned activations and fewer responses to surrounding textile textures, indicating more defect-focused and texture-robust representations.

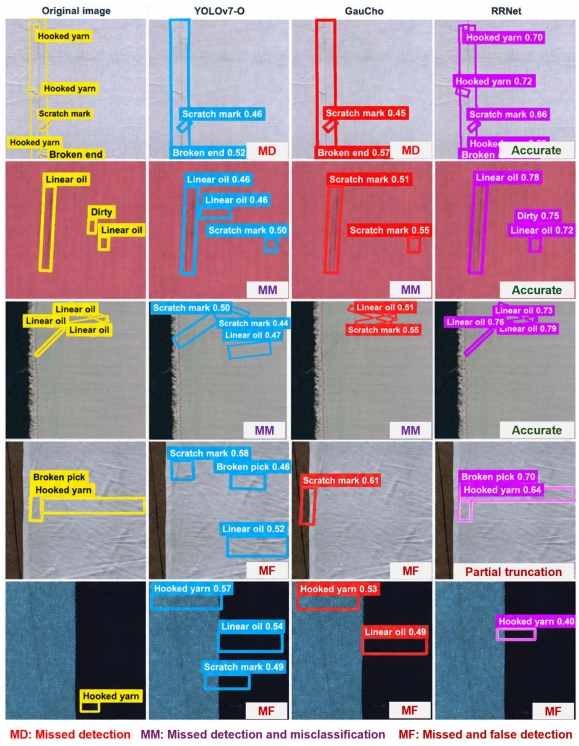


Fig. 4. Qualitative detection comparisons on the IFD dataset under different scenarios. Rows 1–3 show multidefect and cluttered-background scenarios, while Rows 4 and 5 show representative failure cases. Compared with the YOLOv7-O baseline and GauCho, RRNet reduces missed detections and category confusion and achieves more complete rotated localization, although small and low-contrast defects remain challenging.

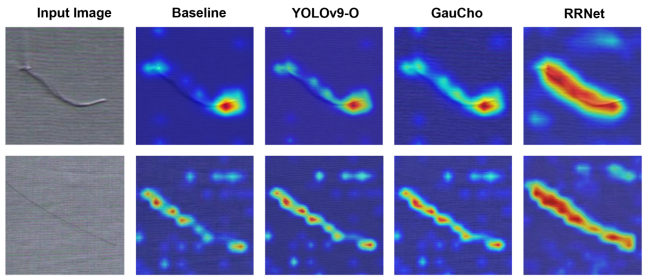


Fig. 5. Grad-CAM visualizations of representative elongated defects. Compared with the baseline, YOLOv9-O, and GauCho, RRNet produces more continuous defect-focused activations while suppressing responses to surrounding textile textures.

VI. CONCLUSION

This article proposes RRNet for deployment-oriented industrial fabric inspection in complex textile scenes. We formulate the task as supervised rotated object detection and introduce two stage-specific modules: PGFM for fusion-stage positional-correspondence recalibration and MSRSC for structure–direction coupled prediction. PGFM reduces texture-induced positional corruption during cross-scale feature fusion, while MSRSC converts the preserved spatial evidence into scale-

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and direction-consistent prediction responses for slender rotated defects. We also construct the IFD dataset with rotated annotations and validate RRNet on both IFD and ZJU-Leaper under a unified OBB protocol.

The results show that RRNet achieves a favorable accuracy–efficiency tradeoff for online fabric inspection. The categorywise analysis further shows the complementary roles of the two modules: PGFM improves robustness to texture interference, while MSRSC provides targeted gains for elongated defects with informative orientation cues. This study focuses on same-distribution deployment on a single production line. Future work will extend RRNet to additional lines with similar imaging hardware and fabric types, conduct cloth-type held-out validation when balanced production data become available, and improve the detection of compact or weakly oriented defects while preserving deployment efficiency.

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